

Uniform convergence of the normalised chart integral on compact input sets

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Abstract

Unit 211 (programme A2, step 3) moves the inputs into the Polish space $E = C([0, 1]^d, \mathbb{R}) \times C([0, 1]^d, \mathbb{R})$ (phase, amplitude; sup metric). A continuous function on the closed cube is extended to $\mathbb{R}^d$ by the coordinatewise clamp $u \mapsto (\max(0, \min(1, u_i)))_i$ (**extCube**: continuous, bounded by the sup norm, the identity on the cube, and $\|\text{ext}f - \text{ext}g\|_\infty \leq d(f, g)$). With $F_N(x) = \text{chartIntegral}(\text{ext } x_1, \text{ext } x_2) / S_N$ and $F(x) = \text{phaseCoeff}(\text{ext } x_1, \text{ext } x_2)$: $F_N \rightarrow F$ pointwise (Headline XIII); F_N ($N \geq N_0(R)$) and F are Lipschitz on the closed ball of radius R in E (units 209–210); and therefore $F_N \rightarrow F$ *uniformly on every compact* $K \subseteq E$ (**tendstoUniformlyOn_normChart**), by the finite-net argument: K lies in a ball, a finite δ -net $T \subseteq K$ exists, pointwise convergence at the net points is eventual, and the two Lipschitz bounds carry it to all of K . This is the deterministic input to the random-phase transfer. Zero sorry/axiom; 9088 jobs.

1 The things formalised

```
instance compactSpace_closedCube (d) : CompactSpace (closedCube d)
def clampCube (d) (x : Fin d → ℝ) : closedCube d := (max 0 (min 1 (x i)))
theorem continuous_clampCube; theorem clampCube_of_mem (hx : x ∈ closedCube d) :
  ↑(clampCube d x) = x
noncomputable def extCube (f : C(closedCube d, ℝ)) : (Fin d → ℝ) → ℝ := fun x => f
  (clampCube d x)
theorem continuous_extCube; theorem abs_extCube_le : |extCube f x| ≤ ||f||
theorem abs_extCube_sub_le : |extCube f x - extCube g x| ≤ dist f g
abbrev InputSpace (d) := C(closedCube d, ℝ) × C(closedCube d, ℝ)
noncomputable def normChart (h k) (l N) (x : InputSpace d) : ℝ :=
  chartIntegral d h k N (extCube x.1) (extCube x.2) / leadScale h k l N
noncomputable def limChart (h k) (l N) (x : InputSpace d) : ℝ := phaseCoeff h k l
  (extCube x.1) (extCube x.2)
theorem tendsto_normChart ... (x) : Tendsto (fun N => normChart h k l N x) atTop (
  (limChart h k l N x))
theorem normChart_lipschitz_eventually ... (R) : L, 0 L N in atTop, x y
  closedBall 0 R,
  |normChart h k l N x - normChart h k l N y| ≤ L * dist x y
theorem limChart_lipschitz ... (R) : L, 0 L x y closedBall 0 R,
  |limChart h k l N x - limChart h k l N y| ≤ L * dist x y
theorem tendstoUniformlyOn_normChart ... (K) (hK : IsCompact K) :
  TendstoUniformlyOn (fun N x => normChart h k l N x) (limChart h k l N) atTop K
```

2 Mathematics

Given ε , put $\delta = \varepsilon/(3(L + L' + 1))$ with L, L' the Lipschitz constants on the ball containing K ; `finite_cover_balls_of_compact` gives a finite $T \subseteq K$ with $K \subseteq \bigcup_{t \in T} B(t, \delta)$; for N large (Lipschitz threshold and the finitely many pointwise convergences, `Filter.eventually_all_finite`), and $x \in K$ with $d(x, t) < \delta$: $|F(x) - F_N(x)| \leq L'\delta + \varepsilon/3 + L\delta < \varepsilon$. Taking the net inside K matters: the Lipschitz bounds are only available on the ball containing K .

3 Lean notes

- `C(closedCube d, )` gets its sup norm and metric only once `CompactSpace (closedCube d)` is an instance; register it from `isCompact_iff_compactSpace` before anything else.
- `Metric.totallyBounded_iff'` (net inside the set) is not a Mathlib name; `finite_cover_balls_of_compact hK h : t s, t.Finite s x t, ball x e` is.
- `filter_upwards` already unfolds `Metric.ball` to `dist ... <`; a following `rw [Metric.mem_ball]` fails.
- Product metric facts: `Prod.dist_eq (max)`, `Prod.norm_def (max)`, `mem_closedBall_zero_iff`.

4 Next

Unit 212: random inputs $X_n \Rightarrow Z$ in E with $N_n \rightarrow \infty$: show $F_{N_n}(X_n) - F(X_n) \rightarrow 0$ in probability (uniform convergence on a compact K carrying most of the limit law, extended to its δ -thickening by the Lipschitz bounds, plus portmanteau for the open thickening), then $F(X_n) \Rightarrow F(Z)$ by continuous mapping, and a Slutsky-type lemma to conclude $F_{N_n}(X_n) \Rightarrow F(Z)$.